

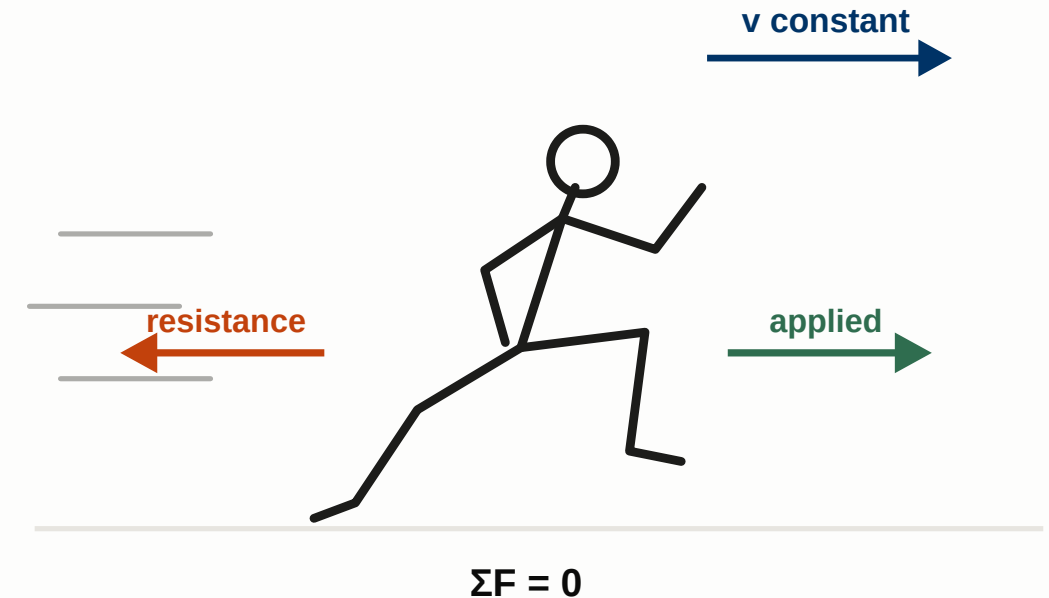
BIOMECHANICS (EPHE 341)

Dynamic Analysis

Last week every equation had zero on the right-hand side. This week the zeros become ma and $I\alpha$, and the method does not change.

Dynamic Equilibrium

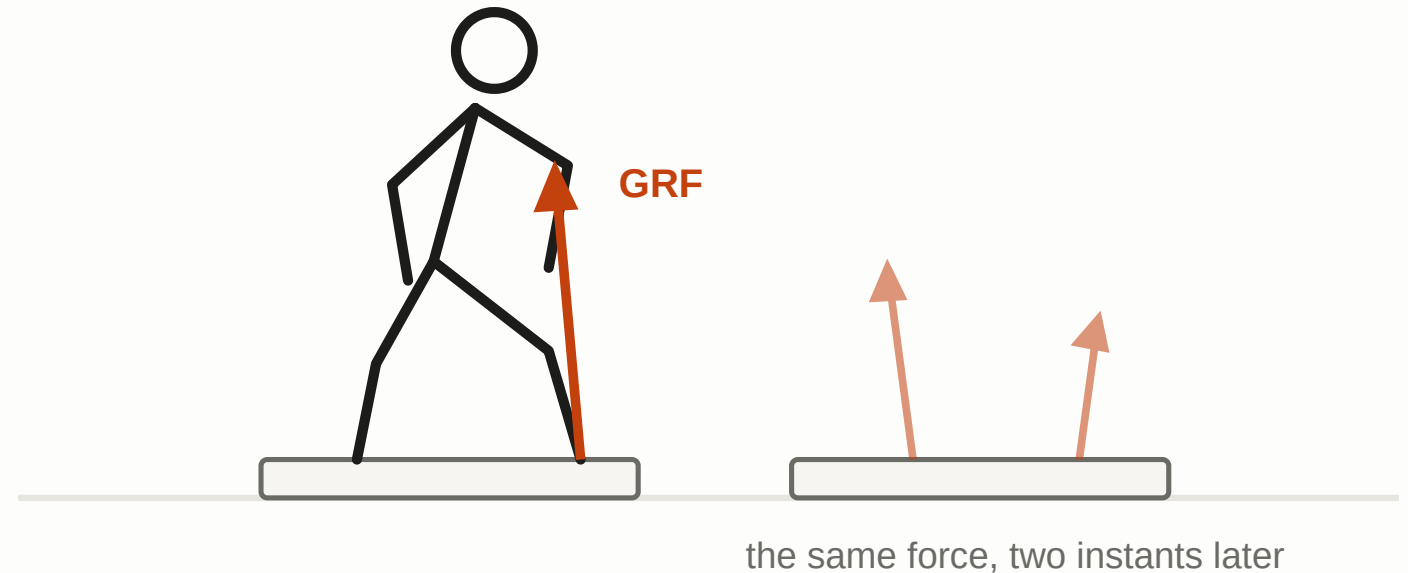
- State of an object which is moving with constant* (uniform) linear and/or angular velocity. * no acceleration.
- Dynamic equilibrium exists if there is a balance between the applied forces and inertial forces for a body in motion, with all the applied forces resulting in equal and oppositely directed inertial forces.
- Therefore the same rules apply from static to dynamic equilibrium but you need to keep in mind that the object is in motion.



Moving quickly is not the same as accelerating. Equal and oppositely directed, so the sums are still zero.

When it is not in equilibrium

- When an object is not in equilibrium the net forces do not equal zero and it requires a **non-equilibrium dynamic analysis**.
- To understand dynamic analysis we need to start with **Newton's second law**.



A plate measures the one external force we can see. Nothing in this picture sums to zero.

Newton's Second Law — the law of acceleration

LINEAR

A net force applied to a body causes an acceleration of that body of a magnitude proportional to the force, in the direction of the force, and inversely proportional to the body's mass.

$$\Sigma F = ma$$

ANGULAR

A net torque (moment) applied to a body causes an angular acceleration of that body of a magnitude proportional to the torque, in the direction of the torque (moment), and inversely proportional to the moment of inertia.

$$\Sigma M = I\alpha$$

Two cautions that cost marks later. The a in $F = ma$ is the acceleration of the **centre of mass** — not of the heel, not of the joint. And the M and the I in $M = I\alpha$ must be taken about the **same axis**.

In two dimensions

- In a 2-dimensional analysis it is important to consider the components of forces and moment(s).
- This leaves us with the **3 equations** of a system under dynamic conditions.

$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$

$$\Sigma M = I\alpha$$

Leonhard Euler (1707–1783) wrote the rotational law for a rigid body, and the two laws together describe any motion of anything solid. Hence the **Newton–Euler** equations, which is what this method is called in every paper that uses it.

Three equations means **three unknowns**, and not one more. Everything about how inverse dynamics is organised follows from that one sentence.

Why two dimensions

- Dynamic analysis of a multi-segment rigid body (like a human body) in 3-dimensions can be very complex.
- We will cover **2-dimensional** examples of dynamic analysis.

Everything from here is planar: motion in the sagittal plane, forces in the sagittal plane, one moment axis perpendicular to it.

This is not only a teaching simplification. A large fraction of published gait analysis reports sagittal joint moments computed exactly this way, because the sagittal plane is where most of walking happens.

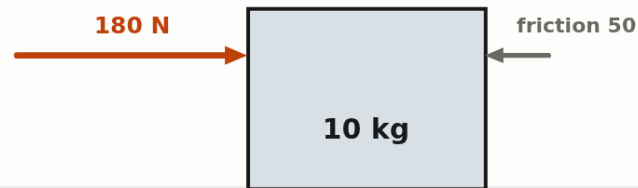
At the end of the lecture we run the method on a real stride and compare it with a three-dimensional solution of the same data.

The same free body, with and without acceleration

$$\Sigma F_x = m \cdot a_x$$

$m \cdot a = 130 \text{ N}$

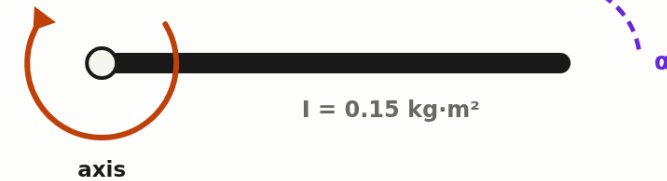




$$180 - 50 = 10 \cdot a$$
$$a = 13.00 \text{ m/s}^2$$

$$\Sigma M = I \cdot \alpha$$

$M = 12 \text{ N}\cdot\text{m}$



$$12 = 0.15 \cdot \alpha$$
$$\alpha = 80.0 \text{ rad/s}^2$$

The free body diagram does not change · the force table does not change · the sign conventions do not change · the only thing that changes is that the right-hand side is no longer zero

The push

180 N

The twist

12 N·m

Last week the right-hand side was zero and you solved for the forces that made it so. This week it is $m \cdot a$ and $I \cdot \alpha$, and you solve for the forces that produce the motion you measured. Slide the push down to 50 N, where it exactly cancels friction, and the dynamic problem turns back into the static one without a single line of the method changing. The dashed arrow is the quantity $m \cdot a$ drawn as though it were a force: add it and the sums are zero again, which is why we call this **dynamic equilibrium**.

Two directions

Forces in, motion out — or motion in, forces out. One of them you can do to a person.

Forward dynamics

- There are **2 approaches** to solving dynamic problems depending on what variables are known or can be manipulated.
- Forward and inverse dynamics.

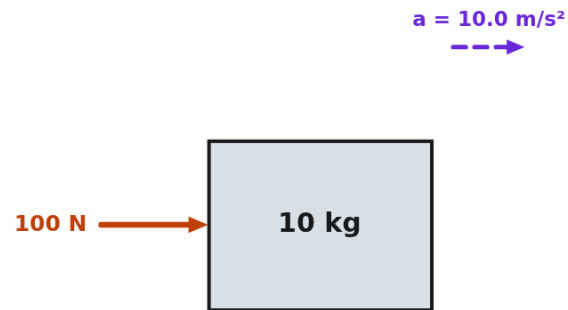
FORWARD DYNAMICS ANALYSIS

- When you know the forces and are trying to determine the accelerations.
- An example of this is in robotics where you have motors that produce a prescribed force or torque.
- That is, given the forces and torques, what are the linear and angular accelerations?

forces $\rightarrow a, \alpha \rightarrow$ motion

This is the direction every physics problem you have ever been set runs in. It is also the direction you **cannot** run on a person, because nobody can tell you what the muscle forces were.

Example problems, forward dynamics



A force of 100 N on a 10 kg object. What is the acceleration?

$$\Sigma F = m \cdot a$$

the forces all act through the centre of gravity, so nothing turns

$$\Sigma F = 100 = 100 \text{ N}$$

$$a = 100 / 10 = 10.00 \text{ m/s}^2$$

the slide too

Forward dynamics: the forces are given and the motion comes out. This is the direction a simulation runs, and the direction you can almost never use on a person, because the forces inside are the part you cannot measure.

#1

#2

#3

#4

Four problems, one equation each, and between them they are the whole of forward dynamics: **known forces in, motion out**. Two linear, two angular. Nothing here needs a free body diagram with unknowns in it, which is exactly what makes the inverse problem — the rest of this lecture — harder.

Simulation

Another application of forward dynamics is in the **simulation of human movement**, when approximate muscle forces are input into models to replicate the performance of a movement.

Build a model — segments, joints, muscles with force–length and force–velocity properties — choose a pattern of activation, and let the equations run. The model walks, or jumps, or falls over.

Then change something you could never change in a person: delete a muscle, weaken a tendon, alter a moment arm, add a prosthesis. Run it again.

Forward dynamics answers *what would happen if*. Inverse dynamics answers *what did happen*. We need both, and they are not substitutes.

Inverse dynamics analysis

- When you know the **kinematics** and are trying to solve for the **kinetics**.
- An example of this type of solution is when you can measure the movement of the body segment and want to solve for the muscle force and joint reaction forces required to produce that movement.

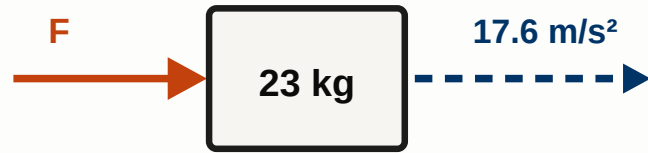
motion + external forces → internal forces

You cannot put a load cell inside somebody. You can put a camera in front of them and a force plate under them — so for human movement this is the only direction available.

Everything from here to the end of the lecture is this one calculation.

Example Problem #1 (inverse dynamics)

What is the resultant force applied to a 23 kg object if the acceleration is 17.6 m/s²?



SOLUTION

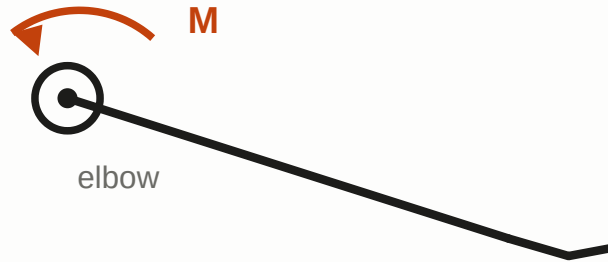
$$\Sigma F_x = m \cdot a_x$$

$$F = 23(17.6)$$

$$F = 404.8 \text{ N}$$

Example Problem #1 (inverse dynamics)

Calculate the moment required for an angular acceleration of 112 rad/s^2 , given the moment of inertia about the elbow is $0.18 \text{ kg}\cdot\text{m}^2$.



SOLUTION

$$\Sigma M = I \cdot \alpha$$

$$M = 0.18(112)$$

$$M = 20.2 \text{ N}\cdot\text{m}$$

The method

Break the limb into segments, start at the one touching the ground, and carry the answer upward.

Inverse dynamics in human gait analysis

- We can record the movement of the human body using a motion capture system, video camera, accelerometers, gyros, etc...
- and measure the external forces,
- but we do not know what is going on **inside** the body to produce the motion.

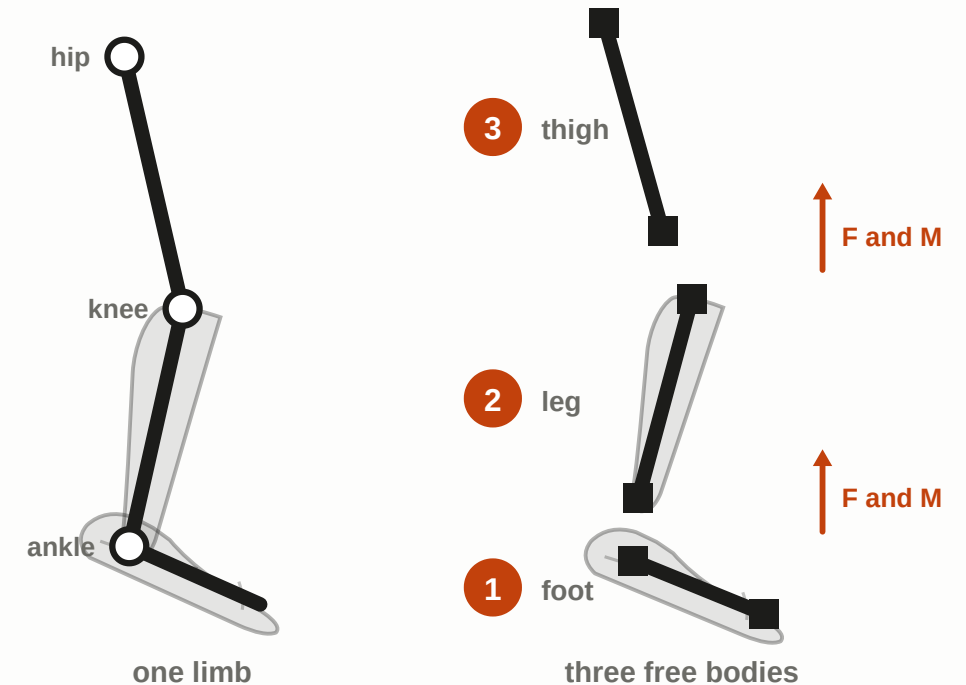
Based on recorded data (motion trajectories, GRFs) and simple dynamics principles (Newton–Euler equations) we can reconstruct the **net joint moments and forces**.

The word **net** is doing a great deal of work in that sentence, and we come back to it before the end.

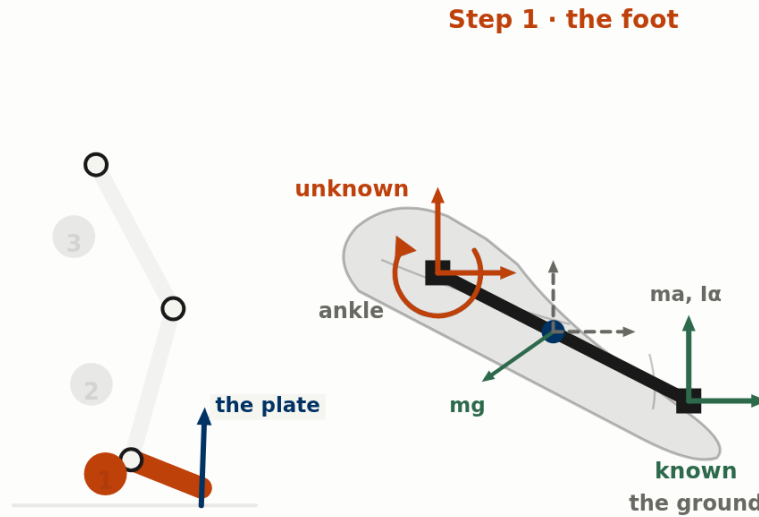
One segment at a time, starting at the bottom

- Inverse dynamics involves taking into account all the forces and moments that act on each segment, and the effect of these forces and moments on the following segments.
- For example, if we want to find out the forces and moments at the hip during a movement we need to consider **each segment separately**.
- We need to start with the **most distal segment** and then carry the solutions for each joint forward.

Step 1: Foot · **Step 2:** Leg · **Step 3:** Thigh



Why the order is forced



What arrives at the distal end

- GRFx, GRFy — measured by the plate
- where it acts — the centre of pressure
- no moment: the ground pushes, it does not twist

What the anthropometric table gives

- m, l and the centre of mass of the foot, and a and α from the markers

What is unknown — three things

- Fankle-x Fankle-y Mankle

Three equations close it

$$\Sigma F_x = m a_x \quad \Sigma F_y = m a_y \quad \Sigma M = I \alpha$$

3 unknowns, 3 equations — solvable. Then the ankle force and the ankle moment go up to step 2.

A free body in two dimensions gives three equations. An unknown joint costs three unknowns. So a segment can be solved only when one of its ends is already known — and at the start exactly one end of one segment is.

Step 1 · the foot

Step 2 · the leg

Step 3 · the thigh

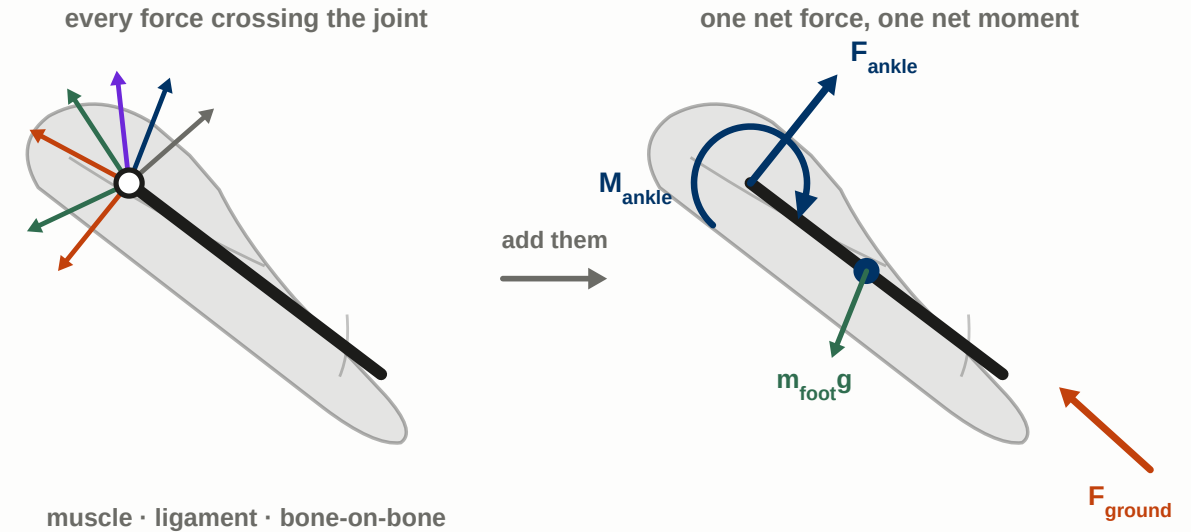
hide the limb

Three segments, three solutions, each one handing the next the force and the moment it needs. The **foot** can go first because a force plate has already measured what happens at its far end, and because the ground applies no moment. Everything above it is the same three equations with the previous answer arriving, reversed, at the bottom.

A net joint force and a net joint moment

1. We need to consider all the forces acting upon the segment and the moments that they would produce — bone-on-bone forces, ligament forces, every muscle crossing the joint and the joint moment of force it produces.
2. Really we can say that we can add all these forces to create a **net joint force** and a **net joint moment**, and then account for the internal and external moments.

One force and one moment, at **each joint, for each segment**. That replacement is exact for the mechanics — the segment accelerates identically — and lossy for the anatomy, because afterwards there is no way of telling which muscle did it.



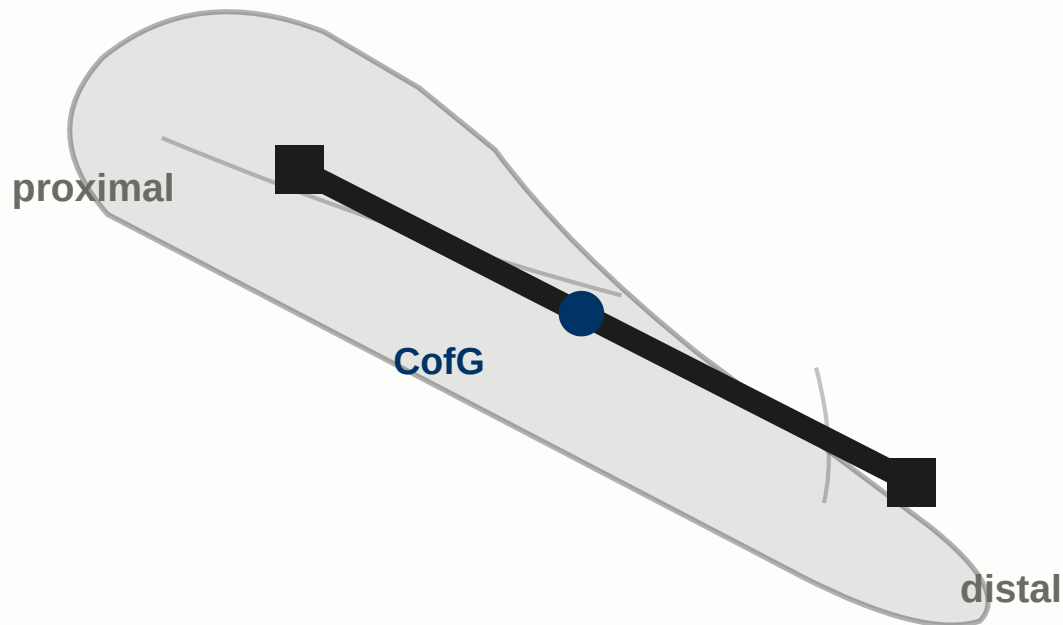
A really simple diagram

So let's make a really simple diagram so we can see the steps necessary to calculate the net joint forces and moments.

$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$

$$\Sigma M = I\alpha$$



NB. The technique for calculating the net joint forces and moments is slightly different from the technique we are used to. It is the same in principle, but it may feel a bit different because we are taking moments about a different axis. Either technique could be used, but the one I am showing you simplifies the calculations given the number of variables.

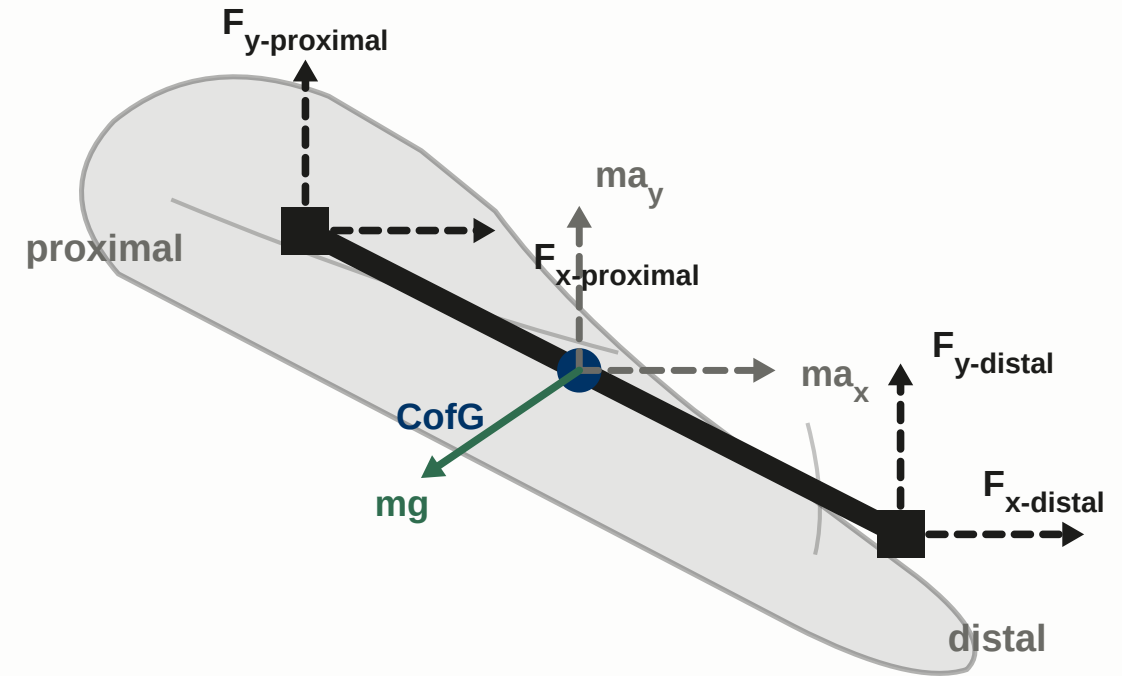
The forces, in general form

If we calculate all values with reference to the **centre of mass** we can use the general forms of the Newton–Euler equations that we learned earlier.

$$\Sigma F_x = F_{x\text{-proximal}} + F_{x\text{-distal}} = ma_x$$

$$\Sigma F_y = F_{y\text{-proximal}} + F_{y\text{-distal}} + mg = ma_y$$

Hover any term to find it on the diagram. And while these are the general forms of the equations you need to keep close attention to the **directions (signs)** of the forces when calculating the values.



The forces about the **distal** end are the ground reaction forces. The forces about the **proximal** end are normally the unknown forces.

The forces, with specific values

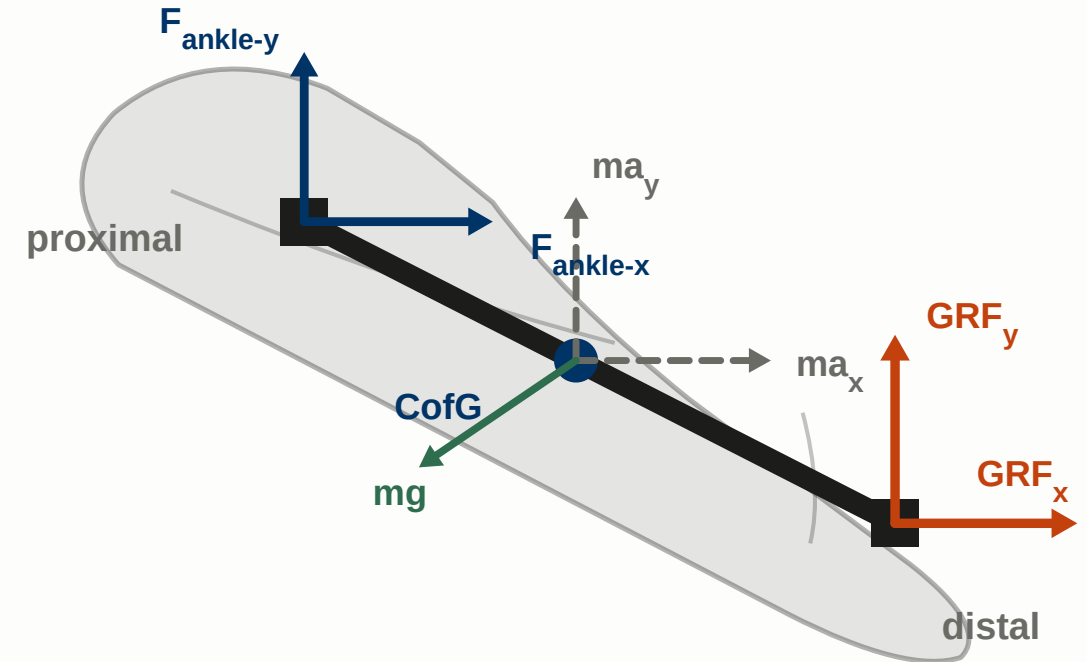
In this example we can give specific forces.

$$\Sigma F_x = F_{\text{ankle-x}} + \text{GRF}_x = ma_x$$

$$F_{\text{ankle-x}} = ma_x - \text{GRF}_x$$

$$\Sigma F_y = F_{\text{ankle-y}} + \text{GRF}_y + mg = ma_y$$

$$F_{\text{ankle-y}} = ma_y - \text{GRF}_y - mg$$



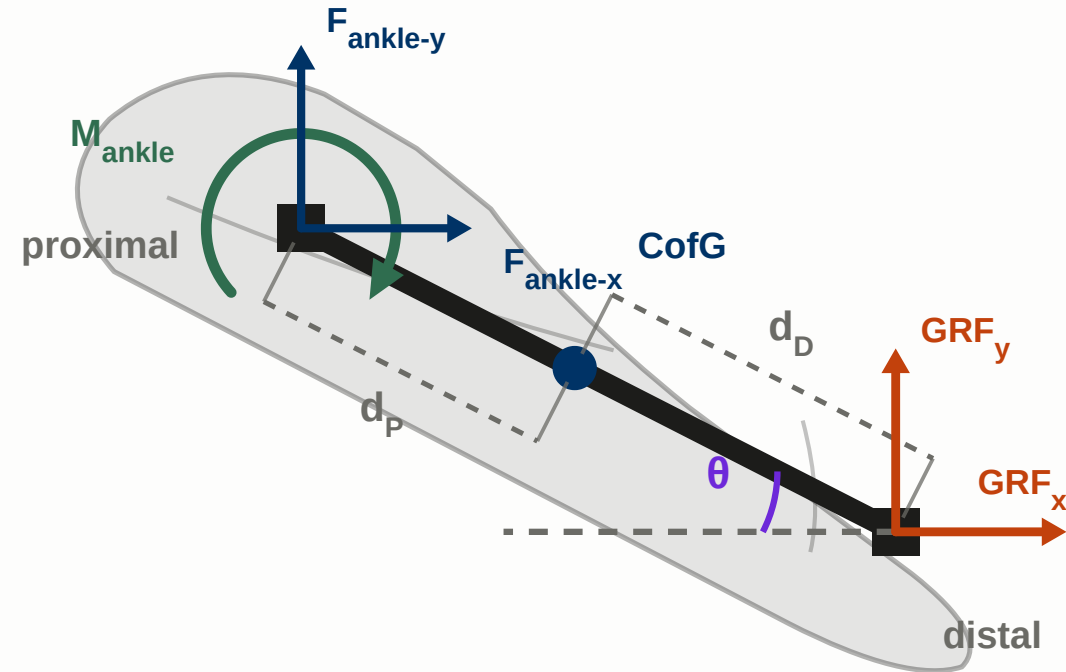
The muscle force acting at the ankle · the ground reaction force

The moments

$$\Sigma M = I\alpha$$

$$M_{\text{ankle}} + M_{\text{Fankle/CG}} + M_{\text{GRF/CG}} = I\alpha$$

$$\begin{aligned} &M_{\text{ankle}} \\ &+ F_{\text{ankle-x}} \cdot d_P(\sin\theta) \\ &+ F_{\text{ankle-y}} \cdot d_P(\cos\theta) \\ &+ \text{GRF}_x \cdot d_D(\sin\theta) \\ &+ \text{GRF}_y \cdot d_D(\cos\theta) \\ &= I\alpha \end{aligned}$$



The same moments, with the geometry drawn

$$\Sigma M = I\alpha$$

M_{ankle}

$$+ F_{\text{ankle-x}} \cdot d_P(\sin\theta)$$

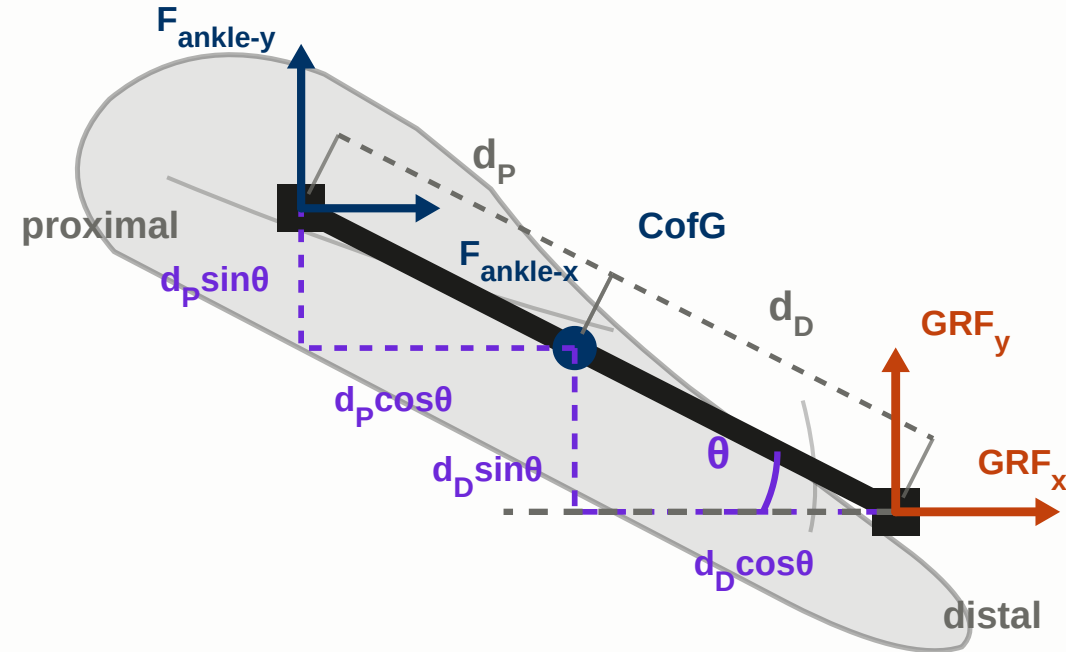
$$+ F_{\text{ankle-y}} \cdot d_P(\cos\theta)$$

$$+ GRF_x \cdot d_D(\sin\theta)$$

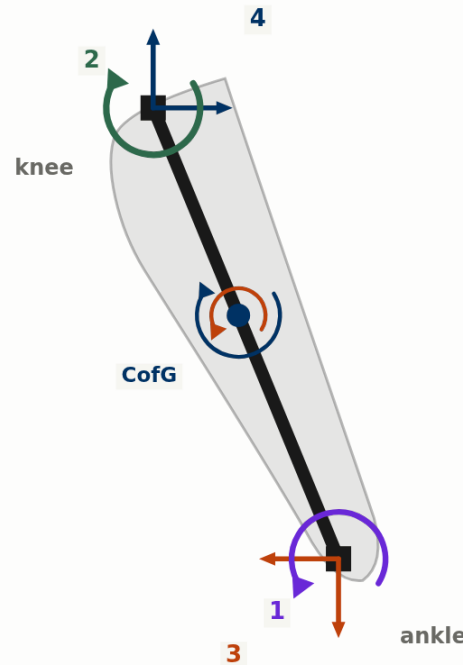
$$+ GRF_y \cdot d_D(\cos\theta)$$

$$= I\alpha$$

$d_P \sin\theta$ and $d_D \sin\theta$ are **vertical** offsets, and they go with the **horizontal** forces. The arm a force turns through is perpendicular to it.



Every segment carries four moments



Four moments, and they add to $I\alpha$

● -M ankle the moment applied at the far end	-0.394 N·m
● M knee the moment at the joint we are solving	1.716 N·m the unknown
● M Fankle/CG the far-end force, about the CofG	-1.094 N·m
● M Fknee/CG the joint force, about the CofG	2.141 N·m

sum of the four **2.369 = $I\alpha$ = 2.369 N·m**

Four terms, four moments. Term 1 came up from the foot; term 2 is what goes up to the thigh.

The foot, in swing

The foot, in stance

The leg

hide the limb

All four

1 · moment at the far end

2 · moment at the joint

3 · far-end force about the CofG

4 · joint force about the CofG

Every segment has **four** moments acting on it and they add to $I\alpha$: one at each end, and one about the centre of gravity from each end's force. **Term 2 is always the unknown** — the moment at the proximal end is what we are solving for, and it becomes term 1 of the segment above.

Up the chain

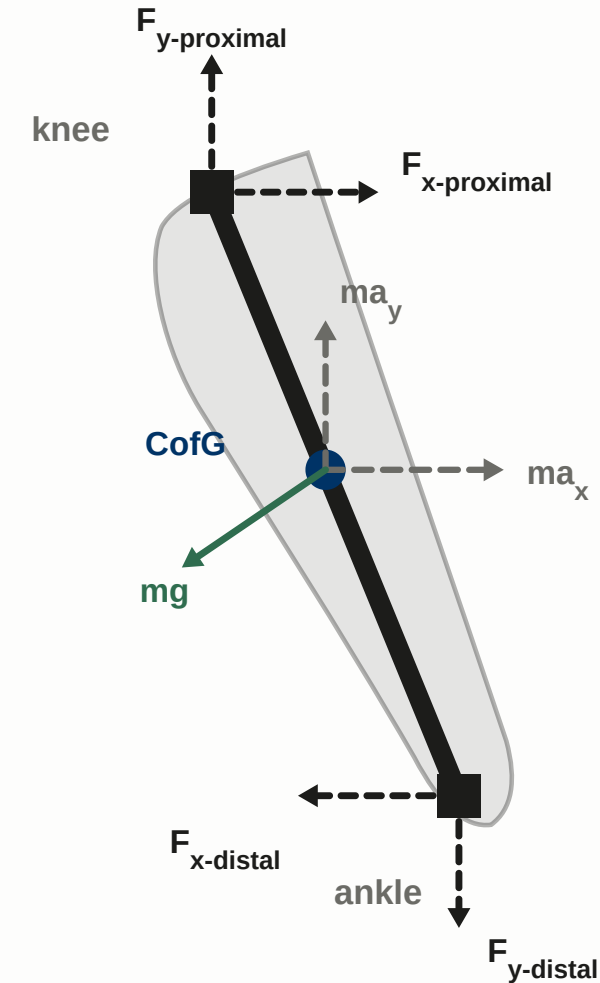
Now that we have found the net ankle joint force and moment we can then use that information to input into the **next limb segment**.

$$\Sigma F_x = ma_x$$

$$\Sigma F_y = ma_y$$

$$\Sigma M = I\alpha$$

Same three equations, new segment. The leg's distal end is the ankle, and we have just finished measuring it.



The distal arrows point the other way now — that is Newton's third law, and it is the next slide.

The leg, with specific forces

In this example we can give specific forces.

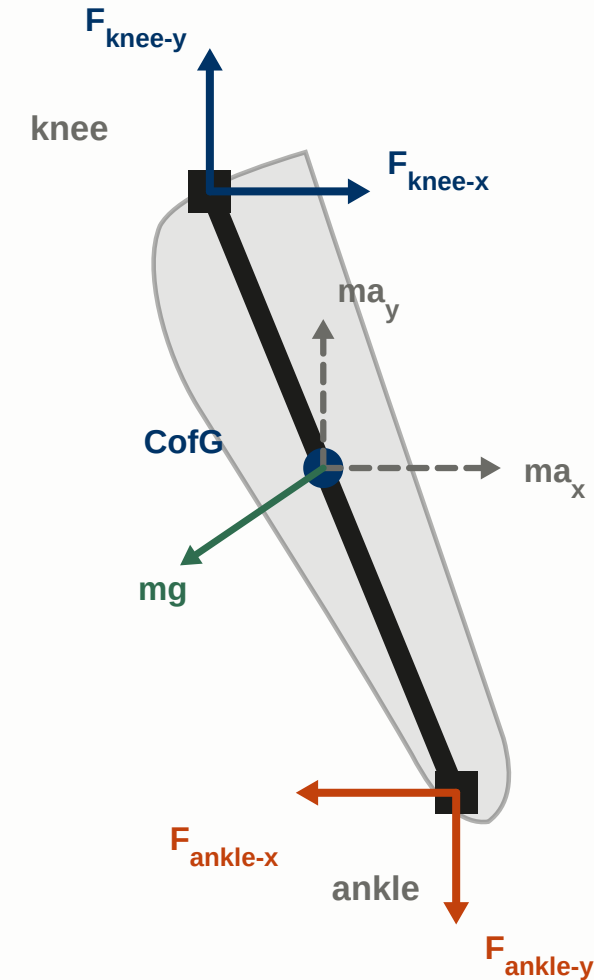
$$\Sigma F_x = F_{\text{knee-x}} - F_{\text{ankle-x}} = ma_{x(\text{leg})}$$

$$F_{\text{knee-x}} = ma_x + F_{\text{ankle-x}}$$

$$\Sigma F_y = F_{\text{knee-y}} - F_{\text{ankle-y}} + mg = ma_{y(\text{leg})}$$

$$F_{\text{knee-y}} = ma_y + F_{\text{ankle-y}} - mg$$

Need to consider **action/reaction** forces. Those minus signs are the whole content of this slide.



The muscle force acting at the knee · the ankle reaction force

The leg, with specific moments

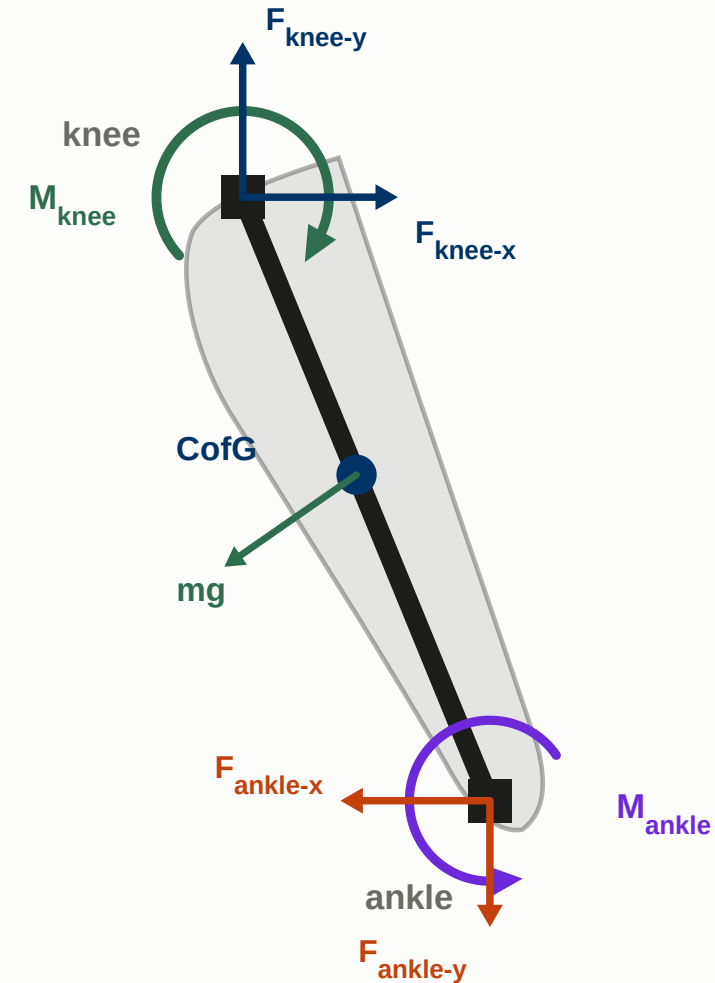
In this example we can give specific moments.

$$\Sigma M = I\alpha$$

$$M_{\text{knee}} + M_{F_{\text{knee}}/\text{CG}} + M_{F_{\text{ankle}}/\text{CG}} - M_{\text{ankle}} = I\alpha$$

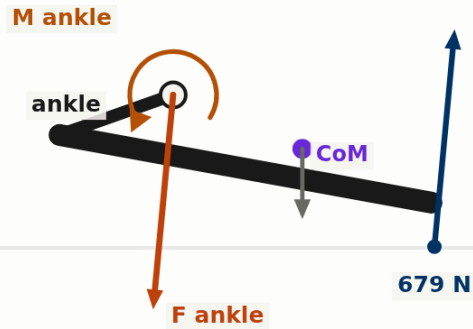
$$M_{\text{knee}} = I\alpha - M_{F_{\text{knee}}/\text{CG}} - M_{F_{\text{ankle}}/\text{CG}} + M_{\text{ankle}}$$

Really just remember that what we solved for at the ankle will affect each limb up the chain.



A moment at **both** ends now — the one we want at the knee, and the one we already solved at the ankle.

The same free body, on a measured stride



44 % of the gait cycle

$$\Sigma F_x = m \cdot a_x$$

$$F(\text{ankle})_x + 63 = 0.80(0.79)$$

$$F(\text{ankle})_x = -62.1 \text{ N}$$

$$\Sigma F_y = m \cdot a_y$$

$$F(\text{ankle})_y + 676 - 7.8 = 0.80(1.33)$$

$$F(\text{ankle})_y = -667.2 \text{ N}$$

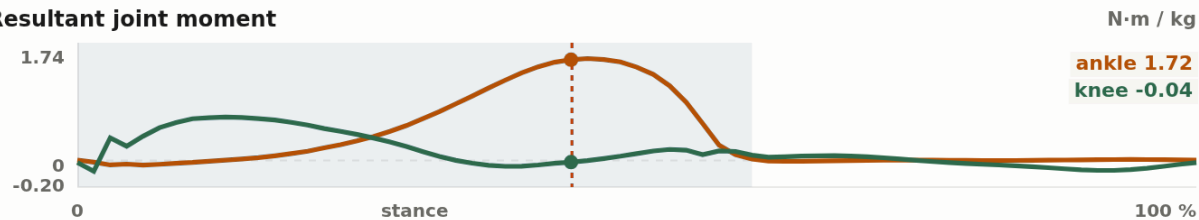
$$\Sigma M = I \cdot \alpha$$

$$M(\text{ankle}) + M(F \text{ ankle}) + M(\text{GRF}) = 0.00381(-25.6)$$

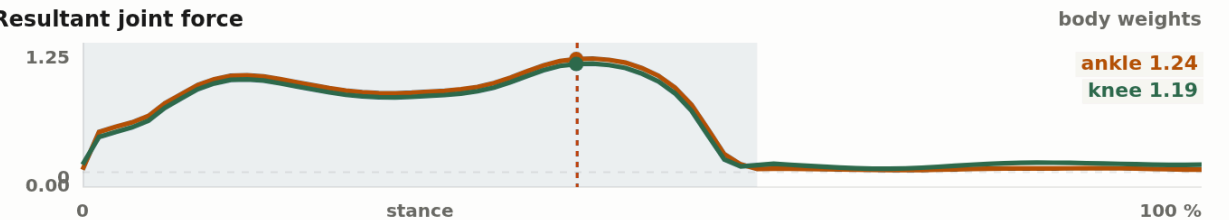
$$M(\text{ankle}) = -94.48 \text{ N} \cdot \text{m}$$

that is 1.718 N·m per kilogram, counting a plantarflexor moment as positive

Resultant joint moment



Resultant joint force



Where in the stride

44 %

The diagram

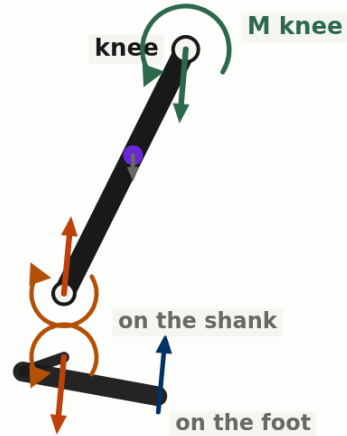
The forces

The moment

▶ Walk

The free body from the last few slides, with a real foot in it, and both answers across the whole stride. Past 60 % the plate loses the foot and the ankle moment collapses to almost nothing.

Handing the ankle to the knee



44 % of the gait cycle

what the foot produced

$$\mathbf{F} = (-62.1, -667.2) \text{ N}, \mathbf{M} = -94.48 \text{ N}\cdot\text{m}$$

reversed onto the shank

$$\mathbf{F} = (62.1, 667.2) \text{ N}, \mathbf{M} = 94.48 \text{ N}\cdot\text{m}$$

$\Sigma F_x, \Sigma F_y$ at the knee

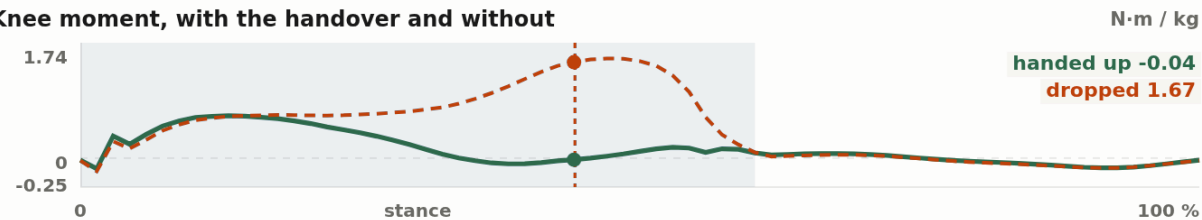
$$\mathbf{F}(\text{knee}) = (-54.1, -638.2) \text{ N}$$

$\Sigma M = I \cdot \alpha$ at the knee

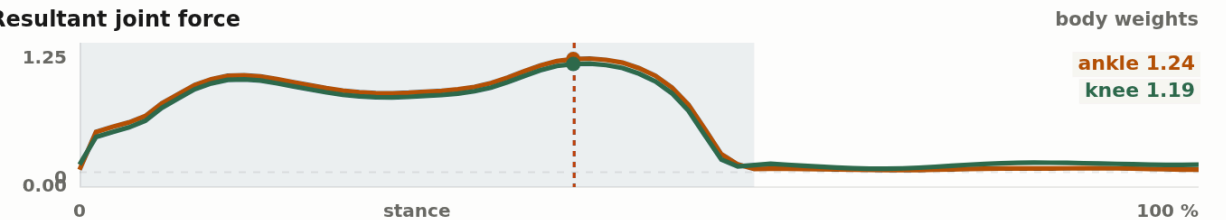
$$\mathbf{M}(\text{knee}) = -2.40 \text{ N}\cdot\text{m} = -0.044 \text{ N}\cdot\text{m/kg}$$

Both halves of the handover are in. At this instant dropping the moment alone would move the knee answer by 94.48 N·m.

Knee moment, with the handover and without



Resultant joint force



Where in the stride

44 %

Hand the moment up

Forget the moment

▶ Walk

Newton's third law applies to moments as well as to forces: the foot's free body produced **two** things at the ankle and **both** come back reversed, as the terms – $\mathbf{M}(\text{ankle})$ and – $\mathbf{F}(\text{ankle})$.

In summary...

What do you need to remember from the last few slides? In order to solve for joint forces and moments:

- You need to know the **kinematics** of the movement and any **external forces**.
- Break up each multi-limb segment into **individual limbs**.
- Use the **Newton–Euler equations** to solve for unknown net forces and moments for each joint, starting from the most distal segment.
- **Transfer** forces and moments up the link of segments.

This gives you an idea of the forces and moments experienced at each joint that result from the external forces and the movements.

What inverse dynamics can and cannot tell us

WHAT IT CAN TELL US

- **NET** joint moments, powers and forces.

WHAT IT CANNOT TELL US

- Contributions/activations of individual muscles.
- Total amount of muscle work done — it does not account for co-contraction of muscles.
- Internal bone-to-bone forces at joints.

OTHER ASSUMPTIONS AND LIMITATIONS

- Frictionless joints.
- Generalized, uniform and/or concentrated mass distribution.
- Estimation of joint centre of rotation.
- Measurement errors: skin and motion artifacts, temporal or spatial misalignment between force plates and motion capture.

The worked example

The ankle and the knee during the swing phase of walking, line by line.

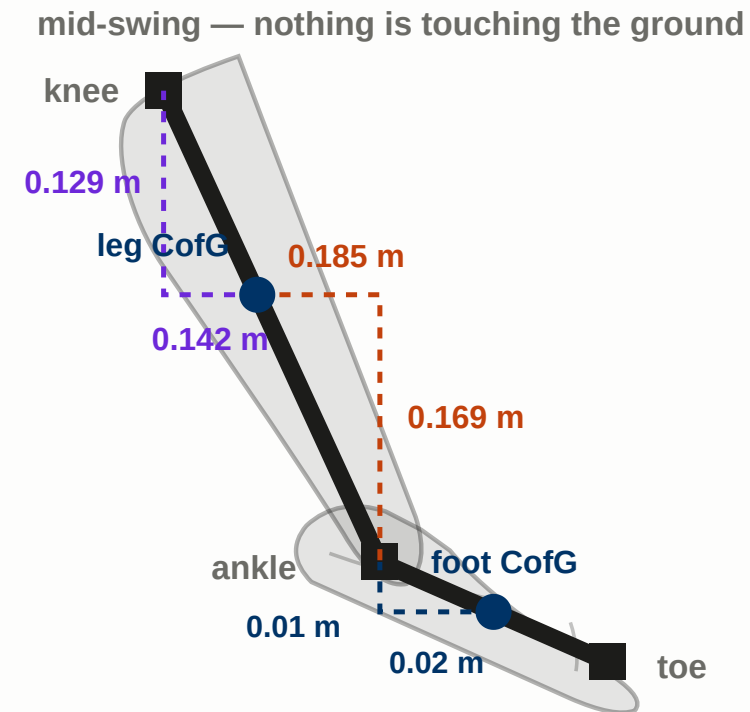
Inverse Dynamics Analysis Example

What you are given

Calculate the forces and moments about the ankle and knee during the **swing phase** of walking using the data below.

	Foot	Leg
m (kg)	1.16	3.72
I (kg·m ²)	0.00995	0.0642
a_x (m/s ²)	9.07	-0.03
a_y (m/s ²)	-6.62	-4.21
α (rad/s ²)	21.63	36.9

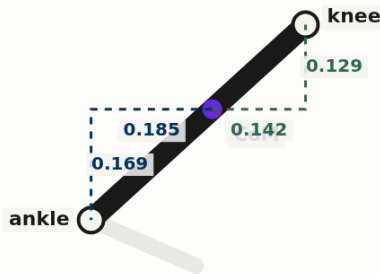
Where did those masses come from?
Winter's table gives the foot 1.45 % of body mass and the leg 4.65 %. $1.16 / 0.0145 =$ **80.0 kg**. $3.72 / 0.0465 =$ **80.0 kg**. The same person, to three figures.



The distances from each joint to each segment's centre of mass come from the same table. Those are the moment arms.

The worked example, every line

the swing phase · slides 30-41



4. the shank, $\Sigma M = I \cdot \alpha$

the knee force about the CoM: 2.141 N·m

the ankle force about the CoM: -1.094 N·m

and the ankle moment, reversed: -0.39 N·m

$$M(\text{knee}) = I \cdot \alpha - (2.14) - (-1.09) - (-0.39) = 1.72 \text{ N}\cdot\text{m}$$

M(F knee/CG)	the slide too
M(F ankle/CG)	the slide too
M(knee)	the slide too

Four steps, two segments, and the whole of the worked example. Solve the foot, reverse what you found, and hand it to the leg.

The swing example

Example 2, in stance

Foot: forces

Foot: moment

Shank: forces

Shank: moment

The same four steps, twice. In **swing** there is no ground force at all, so the ankle moment is nothing but the segment's own inertia and the force at its top end: 0.39 N·m. In **stance** the ground pushes with 450 N a hand's breadth from the centre of mass and the same ankle moment is -76 N·m, two hundred times larger. Nothing about the method changed between them.

On a real stride

The same method, run on measured data, and checked against somebody else's answer.

The recording

One stride of overground walking. A woman, 55 kg, 1.61 m, walking at 1.14 m/s with a cadence of 106 steps a minute and a stride length of 1.31 m.

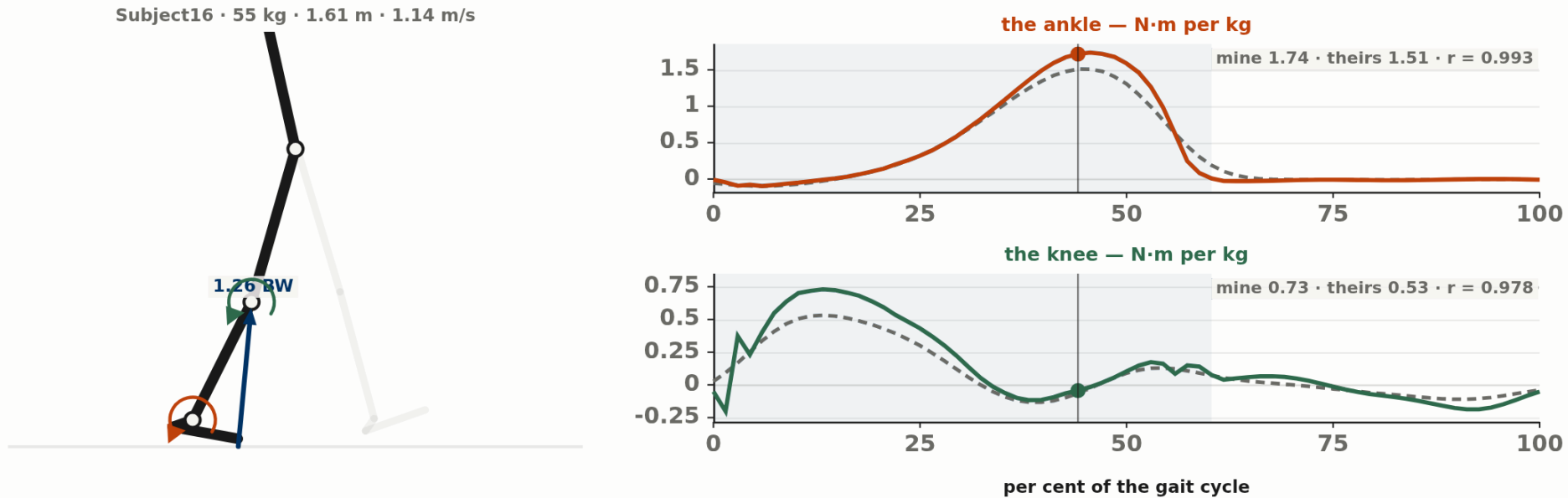
Markers at 60 Hz; the ground reaction force and the centre of pressure at 960 Hz. The plate is under the right foot for the first 60 % of the cycle. Peak vertical force 678 N, which is 1.26 body weights.

Lencioni *et al.* (2019), *Scientific Data* **6**:309, CC BY 4.0 — the same recording we used in the gait analysis and static analysis lectures.

The reason for using this one: the authors **published their own joint moments** for the same stride, computed with their own three-dimensional model. So we can do it from scratch and then mark our own work against somebody who was not trying to agree with us.

Segment masses and inertias from Winter's table, joint centres taken at the skin markers, every derivative taken on a periodically extended record so the seam of the stride does not produce an artefact.

The same method, run for real



Where in the stride

44 %

Both

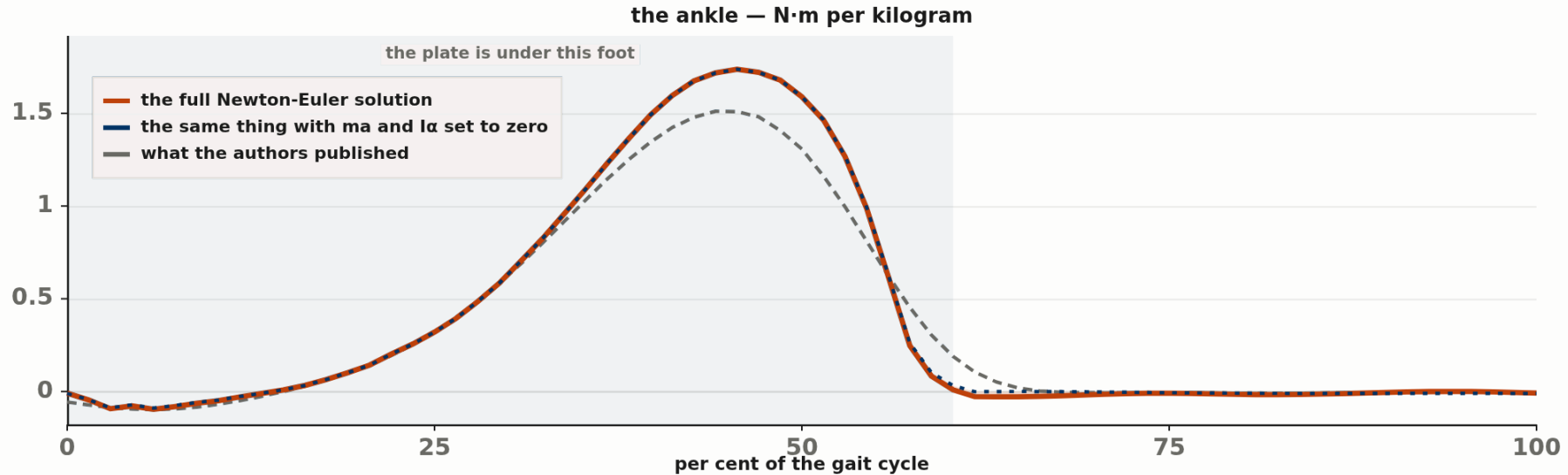
Mine only

Theirs only

▶ Walk

One stride, two answers. The shapes match — $r = 0.99$ at the ankle — and both peaks land on the same instant, while mine run 15 % high at the ankle and 37 % at the knee. No time shift, no centre-of-pressure correction and no single joint-centre nudge closes that gap without making early stance worse: it is the difference between two defensible models of the same body, which is the first three lines of the limitations slide.

When do you actually need the dynamics?



the dynamics are worth 0.06 % here

Over stance the two solutions differ by at most 0.023 N·m/kg. At the push-off peak the full solution gives 1.738 and statics gives 1.737 — a difference of 0.06 %. The foot weighs 0.8 kg and is barely accelerating; the ground is pushing with 678 N a hand's breadth from the ankle. Nothing else could matter.

The ankle

The knee

This is the figure lecture 19 asked for. Last week the walking ankle was solved as a **statics** problem and came out 15 % above the published answer, and the obvious suspect was the missing ma and la . It was not them. Setting every dynamic term to zero moves the ankle's push-off peak by six hundredths of one per cent. The 15 % is the model — whose anthropometric table, and where you decide the joint is. At the knee in swing it is the other way round entirely, and that is the case worth remembering: **no external force, a heavy segment, and dynamics is the whole of the answer.**

What to take away

- Dynamic analysis is static analysis with ma and $I\alpha$ on the right. The diagram, the table and the conventions are unchanged.
- Three equations means three unknowns, which is why you start at the segment touching the ground and work upward.
- Four moments act on every segment, and they add to $I\alpha$: one at each end, and one about the centre of mass from each end's force.
- Both the force **and** the moment reverse at every joint on the way up.
- Take moments about each segment's own centre of mass, and pair each force with the arm **perpendicular** to it.
- A joint moment is a **calculation**, not a measurement, and it is only as good as the anatomy you fed it.

Where this goes next

Multiply a joint moment by the joint's angular velocity and you get **joint power** — how fast that joint is putting energy into the body or taking it out. That one multiplication turns this lecture into the standard clinical description of walking.

And the honest limitation remains: inverse dynamics tells you what the joint did, never which muscle did it. Getting from a net moment to individual muscle forces needs either a measurement we cannot make, or a forward model and an assumption about what the nervous system is trying to optimise.

All done.